§ 1.4 Exercise

44. Suppose that $f(x) = \dfrac{g(x)}{h(x)}$ and $h(a) = 0$. Determine whether each of the following statements is always true, always false or maybe true/maybe false. Explain. (a) $\lim_{x \to a} f(x)$ does not exist. (b) f is not continuous at $x = a$.

(a) X : Let $f(x) = \dfrac{g(x)}{h(x)} = \dfrac{x^2 - a^2}{x - a}$. Then $\lim_{x \to a} f(x) = a + a = 2a$

This implies $\lim_{x \to a} f(x)$ exists.

(b) O : Since $h(a) = 0$, $f(a)$ is not defined.

Then f is not cont. at $x = a$.

47. If f is continuous at $x = a$, prove that $g(x) = |f(x)|$ is continuous at $x = a$.

48. Determine whether the converse of exercise 47 is true. That is, if $|f|$ is continuous at $x = a$, is it necessarily true that f must be continuous at $x = a$?

47. Let $h(x) = |x|$, since h is cont. on $\mathbb{R}$, $h(f(a)) = |f(a)|$ is cont.

Since $\lim_{x \to a} f(x) = f(a)$, we have

$$g(a) = h \circ f(a) = h(f(a)) = |f(a)|$$
$$= \left|\lim_{x \to a} f(x)\right|$$
$$= \lim_{x \to a} |f(x)|$$
$$= \lim_{x \to a} g(x).$$

Hence g is cont. at $x = a$. //

48. Let

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Then $g(x) = |f(x)| = 1$, it is clear that $g = |f|$ is cont. at $x = 0$.

But f is not cont. at $x = 0$, the conclusion is that it is not necessary.

53. Prove that if f is continuous on an interval $[a, b]$, $f(a) > a$ and $f(b) < b$, then f has a fixed point (a solution of $f(x) = x$) in the interval (a, b).

pf Given that $f(a) > a \Rightarrow f(a) - a > 0$

$f(b) < b \Rightarrow f(b) - b < 0$

Let $h(x) = f(x) - x$, since $h(a) = f(a) - a > 0$

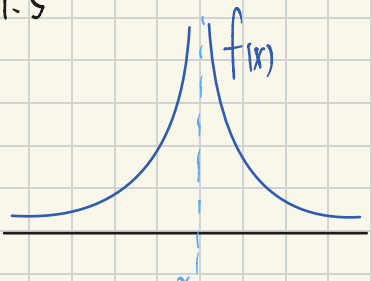
and $h(b) = f(b) - b < 0$,

by the Intermediate Value Theorem, $\exists x \in (a, b)$ st $h(x) = 0$

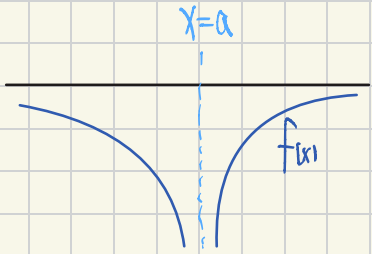
That is $f(x) - x = 0$.

Hence $\exists x \in (a, b)$ st $f(x) = x$. //

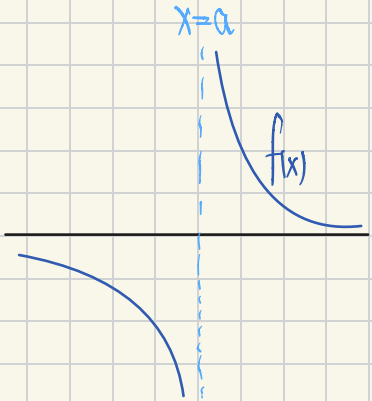
§ 1.5



$\lim_{x\to a} f(x) = \infty$



$\lim_{x\to a} f(x) = -\infty$

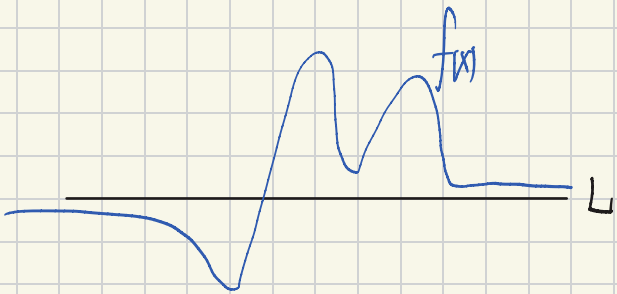


$\lim_{x\to a^-} f(x) = -\infty$

$\lim_{x\to a^+} f(x) = \infty$

$\lim_{x\to a} f(x)$ DNE

$$\begin{cases} \lim_{x\to\infty} f(x) = L \\ \lim_{x\to-\infty} f(x) = L \end{cases}$$



Def

1. $x = a$ is the vertical asymptote of f if $\lim_{x\to a} f(x) = \pm\infty$ or $\lim_{x\to a^+} f(x) = \pm\infty$.
2. $y = L$ is the horizontal asymptote of f if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$.

3. Slant asymptote (Oblique asymptote)

$$m = \lim_{x \to \pm\infty} \frac{f(x)}{x}$$

$$b = \lim_{x \to \pm\infty} [f(x) - mx]$$

$\Rightarrow y = mx + b$ is slant asymptote

§1.7 (老师不太会考)

Evaluate $\lim_{x \to 0} \dfrac{1 - \cos x^2}{x^4}$.

Solution As usual, we look at a graph (see Figure 1.57) and some function values.

x	$\dfrac{1 - \cos x^2}{x^4}$
0.1	0.499996
0.01	0.5
0.001	0.5
0.0001	0.0
0.00001	0.0

x	$\dfrac{1 - \cos x^2}{x^4}$
−0.1	0.499996
−0.01	0.5
−0.001	0.5
−0.0001	0.0
−0.00001	0.0

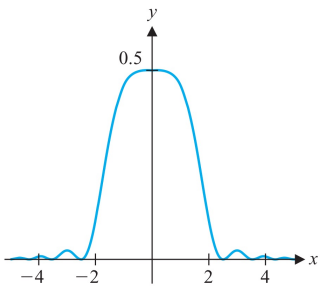


FIGURE 1.57

$$y = \frac{1 - \cos x^2}{x^4}$$

Loss-of-significance errors

$$\frac{1 - \cos x^2}{x^4} = \frac{1 - \cos x^2}{x^4} \cdot \frac{1 + \cos x^2}{1 + \cos x^2}$$ Multiply numerator and denominator by $(1 + \cos x^2)$.

$$= \frac{1 - \cos^2(x^2)}{x^4(1 + \cos x^2)}$$ $1 - \cos^2(x^2) = \sin^2(x^2)$.

$$= \frac{\sin^2(x^2)}{x^4(1 + \cos x^2)}.$$

x	$\dfrac{\sin^2(x^2)}{x^4(1 + \cos x^2)}$
±0.1	0.499996
±0.01	0.4999999996
±0.001	0.5
±0.0001	0.5
±0.00001	0.5

$$\Rightarrow \lim_{x \to 0} \frac{1 - \cos x^2}{x^4} = \lim_{x \to 0} \frac{\sin^2(x^2)}{x^4(1 + \cos x^2)} = \frac{1}{2}$$

§ 1.5

In exercises 1–4, determine (a) $\lim_{x\to a^-} f(x)$, (b) $\lim_{x\to a^+} f(x)$ and (c) $\lim_{x\to a} f(x)$ (answer as appropriate, with a number, ∞, $-\infty$ or does not exist).

4. $f(x) = \dfrac{1-x}{(x+1)^2}$, $a = -1$.

(a) $\lim_{x\to -1^-} f(x) = \infty$

(b) $\lim_{x\to -1^+} f(x) = \infty$

(c) $\lim_{x\to -1} f(x) = \infty$

In exercises 5–20, determine each limit (answer as appropriate, with a number, ∞, $-\infty$ or does not exist).

11. $\lim_{x\to-\infty} \dfrac{-x}{\sqrt{4+x^2}} = \lim_{t\to\infty} \dfrac{t}{\sqrt{4+t^2}} = \lim_{t\to\infty} \dfrac{1}{\sqrt{\frac{4}{t^2}+1}} = 1$ #

Let $t=-x$

$t\to\infty$ as $x\to-\infty$

14. $\lim_{x\to\infty} \dfrac{2x-1}{x^2+4x+1} = \lim_{x\to\infty} \dfrac{\frac{2}{x}-\frac{1}{x^2}}{1+\frac{4}{x}+\frac{1}{x^2}} = \dfrac{0}{1}$

$= 0$ #

17. $\lim_{x\to\infty} \dfrac{3x+\sin x}{4x-\cos 2x} = \lim_{x\to\infty} \dfrac{3+\frac{\sin x}{x}}{4-\frac{\cos 2x}{x}} = \dfrac{3}{4}$ #

In exercises 21–28, determine all horizontal and vertical asymptotes. For each vertical asymptote, determine whether $f(x) \to \infty$ or $f(x) \to -\infty$ on either side of the asymptote.

24. $f(x) = \dfrac{1-x}{x^2+x-2} = \dfrac{-(x-1)}{(x+2)(x-1)}$

① Vertical: Since $\lim_{x \to -2^-} f(x) = \infty$ and $\lim_{x \to -2^+} f(x) = -\infty$,

$x = -2$ is the vertical asymptote.

② Horizontal:

$\lim_{x \to \infty} \dfrac{1-x}{x^2+x-2} = 0$ and $\lim_{x \to -\infty} \dfrac{1-x}{x^2+x-2} = 0$,

$y = 0$ is the horizontal asymptote.

In exercises 29–32, determine all vertical and slant asymptotes.

30. $y = \dfrac{x^2+1}{x-2}$

$= (x+2) + \dfrac{3}{x-2}$

	1	0	1	2
		2	2	
	1	2	3	

Then $y = x+2$ is the slant asymptote.

41. Evaluate $\lim_{x\to\infty}\left(\sqrt{4x^2-2x+1}-2x\right)$

Sol. $\lim_{x\to\infty}\left(\sqrt{4x^2-2x+1}-2x\right)=\lim_{x\to\infty}\left(\sqrt{4x^2-2x+1}-2x\right)\cdot\frac{\sqrt{4x^2-2x+1}+2x}{\sqrt{4x^2-2x+1}+2x}$

$$=\lim_{x\to\infty}\frac{4x^2-2x+1-4x^2}{\sqrt{4x^2-2x+1}+2x}$$

$$=\lim_{x\to\infty}\frac{-2x+1}{\sqrt{4x^2-2x+1}+2x}=\lim_{x\to\infty}\frac{-2+\frac{1}{x}\to 0}{\sqrt{4-\frac{2}{x}+\frac{1}{x^2}\to 0}+2}$$

$$=\frac{-2}{2+2}=\frac{-1}{2}\ \#$$

47. Find a quadratic function $q(x)$ such that $f(x)=\dfrac{x^2-4}{q(x)}$ has one horizontal asymptote $y=-\frac{1}{2}$ and exactly one vertical asymptote $x=3$.

Sol. Since $x=3$ is the vertical asymptote, let $q(x)=a(x-3)^2$

Then $\lim_{x\to\infty}f(x)=\frac{1}{2}\iff\lim_{x\to\infty}\frac{x^2-4}{ax^2-6ax+9a}=\frac{1}{2}$

$$\iff\frac{1}{a}=\frac{1}{-2}$$

$$\iff a=-2$$

Hence $q(x)=-2(x-3)^2$. #

§1.7

4. $\lim_{x\to\infty} x^2\left(\sqrt{x^4+8}-x^2\right) = \lim_{x\to\infty} x^2\left(\sqrt{x^4+8}-x^2\right)\cdot\frac{\sqrt{x^4+8}+x^2}{\sqrt{x^4+8}+x^2}$

$= \lim_{x\to\infty} \frac{x^2\left(x^4+8-x^4\right)}{\sqrt{x^4+8}+x^2} = \lim_{x\to\infty} \frac{8x^2}{\sqrt{x^4+8}+x^2}$

$= \lim_{x\to\infty} \frac{8}{\sqrt{1+\frac{8}{x^4}}+1}$ ($\frac{8}{x^4}\to 0$)

$= \frac{8}{1+1} = \frac{8}{2} = 4$ #

7. $\lim_{x\to 0} \frac{1-\cos 2x}{12x^2} = \lim_{x\to 0} \frac{1-(1-2\sin^2 x)}{12x^2}$

$= \lim_{x\to 0} \frac{2\sin^2 x}{12x^2} = \frac{1}{6}$ #

§ 2.1



slope of secant : $m_{sec} = \dfrac{f(a+h)-f(a)}{(a+h)-a} = \dfrac{f(a+h)-f(a)}{h}$

slope of tangent : $m_{tan} = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h} = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$

DEFINITION 1.2

If $s(t)$ represents the position of an object relative to some fixed location at time t as the object moves along a straight line, then the **instantaneous velocity** at time $t = a$ is given by

$$v(a) = \lim_{h\to 0} \frac{s(a+h)-s(a)}{(a+h)-a} = \lim_{h\to 0} \frac{s(a+h)-s(a)}{h}, \qquad (1.5)$$

provided the limit exists. The **speed** is the absolute value of the velocity.

In exercises 1–8, find the equation of the tangent line to $y = f(x)$ at $x = a$. Graph $y = f(x)$ and the tangent line to verify that you have the correct equation.

2. $f(x) = x^2 - 2$, $a = 0$. $f(0) = -2$

$$m = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{h^2-2+2}{h}$$

$$= \lim_{h\to 0} \frac{h^2}{h} = \lim_{h\to 0} h = 0.$$

So the equation of tangent line is $y + 2 = 0(x-0) = 0$

$\Rightarrow y = -2$ ∗

In exercises 15–18, use the position function $s(t)$ meters to find the velocity at time $t = a$ seconds.

$S(0) = 0$ m

$S(1) = -0.9$ m

16. $S(t) = 4t - 4.9t^2$, (a) $a = 0$ (b) $a = 1$

(a) $v(0) = \lim_{h\to 0} \frac{S(0+h) - S(0)}{h} = \lim_{h\to 0} \frac{4h - 4.9h^2 - 0}{h}$

$= \lim_{h\to 0} 4 - 4.9h = 4$ m/s

(b) $v(1) = \lim_{h\to 0} \frac{S(1+h) - S(1)}{h} = \lim_{h\to 0} \frac{4(1+h) - 4.9(1+h)^2 + 0.9}{h}$ ($1+2h+h^2$)

$= \lim_{h\to 0} \frac{4 + 4h - 4.9 - 9.8h - 4.9h^2 + 0.9}{h}$

$= \lim_{h\to 0} 4 - 9.8 - 4.9h = -5.8$ m/s

36. (a) Show that the graphs of $y = x^2 + 1$ and $y = x$ do not intersect.

(b) Find the value of x such that the tangent lines to $y = x^2 + 1$ and $y = x$ are parallel.

37. (a) Find an equation of the tangent line to $y = x^3 + 3x + 1$ at $x = 1$.

(b) Show that the tangent line in part (a) intersects $y = x^3 + 3x + 1$ at more than one point.

(c) Show that for any number c, the tangent line to $y = x^2 + 1$ at $x = c$ only intersects $y = x^2 + 1$ at one point.

36. (a) $x^2 + 1 = x \Rightarrow x^2 - x + 1 = 0$

Since the discriminant (判別式) is $1^2 - 4 = -3 < 0$, the quadratic equation does not have real solution of x.

Hence $y = x^2 + 1$ and $y = x$ do not intersect. //

(b) The slope of $y = x$ is 1. Then

$\lim_{h\to 0} \frac{(x+h)^2 + 1 - x^2 - 1}{h} = \lim_{h\to 0} \frac{(2x+h)\cdot h}{h} = \lim_{h\to 0} 2x + h = 2x$

Since $2x = 1$, $x = \frac{1}{2}$. #

37. (a) $m = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0} \frac{(1+h)^3+3(1+h)+1-5}{h}$

$= \lim_{h\to 0} \frac{1+3h+3h^2+h^3+3+3h-4}{h}$

$= \lim_{h\to 0} \frac{h^3+3h^2+6h}{h} = \lim_{h\to 0} h^2+3h+6 = 6$

$y(1) = 5$

$\Rightarrow y-5 = 6(x-1)$

$\Rightarrow y = 6x-1$ #

(b) $6x-1 = x^3+3x+1 \Leftrightarrow x^3-3x+2=0$

$\Leftrightarrow (x-1)(x^2+x-2)=0$

$\Leftrightarrow (x-1)(x+2)(x-1)=0$

$\Leftrightarrow x=1,\ x=-2$

Hence $y=6x-1$ and $y=x^3+3x+1$ intersect at more than one point. //

(c) For $c_1, c_2 \in \mathbb{R}$, let $y(c_1) = y(c_2)$.

That $c_1^2+1 = c_2^2+1 \Leftrightarrow c_1^2 = c_2^2$

$\Leftrightarrow c_1 = c_2$

Hence $\forall c \in \mathbb{R}$, $x=c$ intersects $y=x^2+1$ only at one point. //